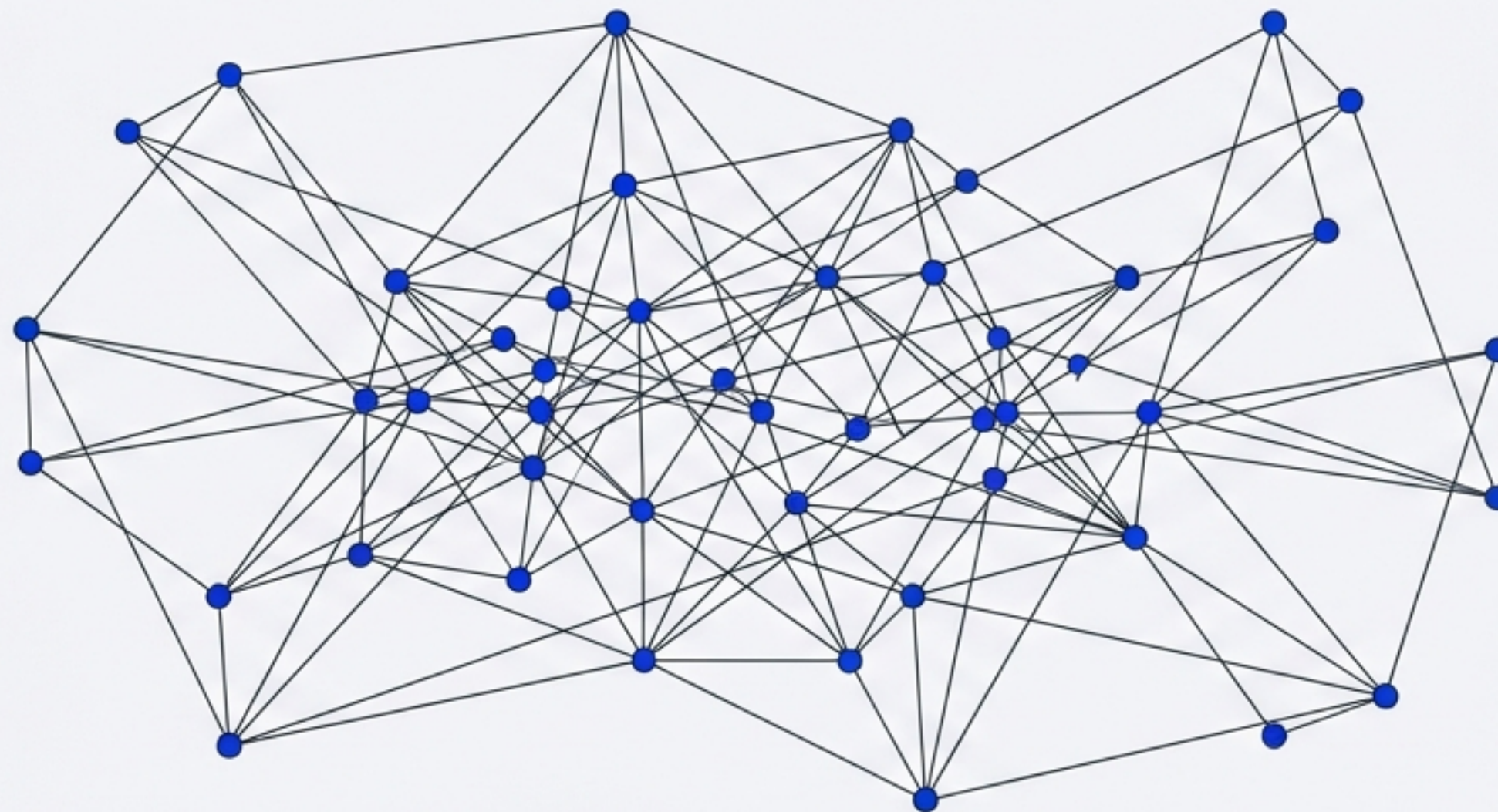


Chain of Trust & Key Graphs

A Decentralised PKI Trust Model for Layered Identity & Personalised Delivery.



Architecture Brief v0.4.12

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Language: EN-GB



DISCOVERY



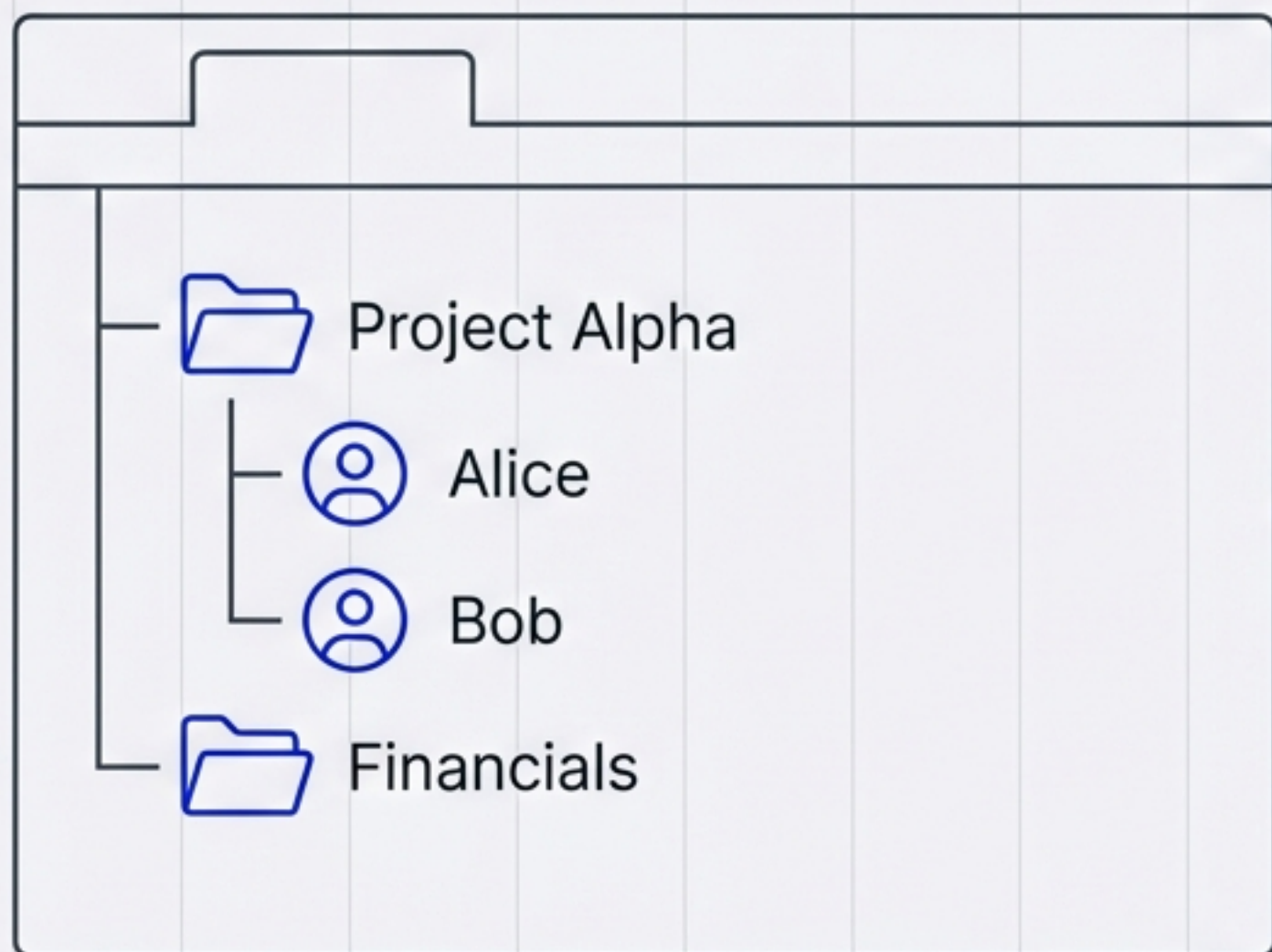
INTEGRITY



DELIVERY

The Server Stores Only Noise

Client Reality



Server Reality

Obj_Id:	a3f7c891
Blob:	0x7f4a2b9...
Token:	UUID-v4
Metadata:	[ENCRYPTED]

Zero Knowledge
Transformation

No strings. No metadata. No injection attacks.

The Problem of Trust Propagation

Scenario: Alice has a key. She needs to message Bob, Carol, and Dave securely.

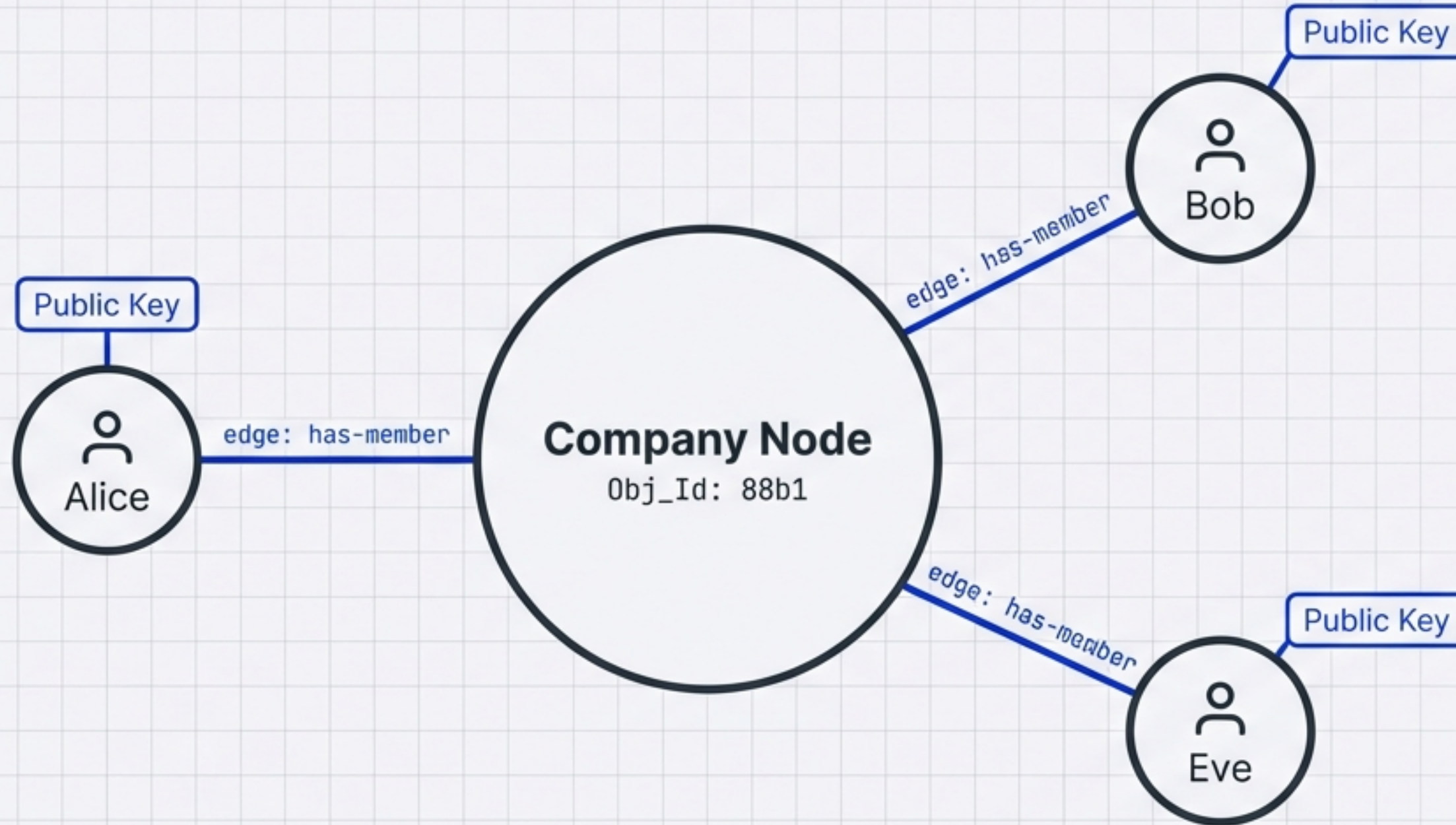
Conflict: We cannot use a centralized directory (metadata leak).

Conflict: Manual exchange of keys does not scale.



From one trusted key, how do we securely discover the entire organisation?

Trust is a Graph, Not a List



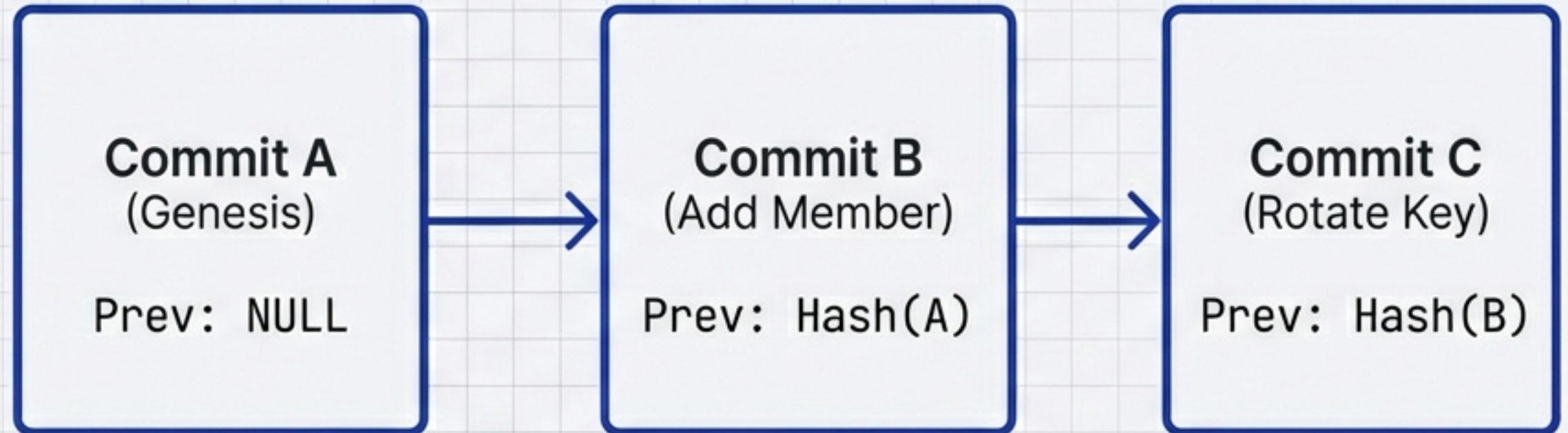
- **Structure:** Native Issues-FS Graph.
- **Traversal:** Client starts at the Trusted Node and walks the edges.
- **Result:** Automatic discovery of all valid keys.

The Key Graph is just another graph in the filesystem. The Cartographer visualises it; AppSec audits it.

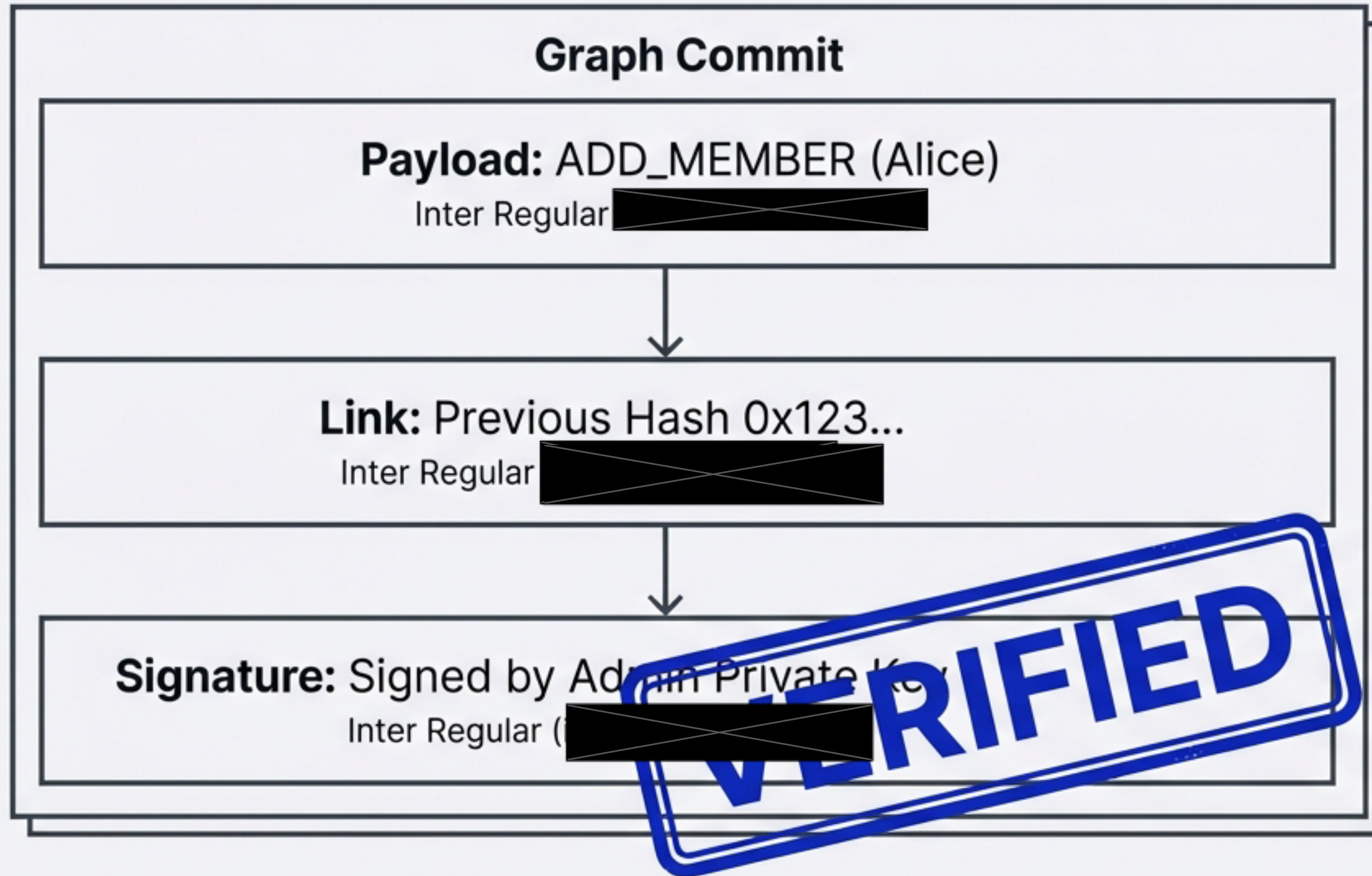
Integrity via Git-Style Hash Chaining

- **Tamper Evidence:** History cannot be rewritten.
- **Append-Only:** Deletion is a REVOKE commit.
- **Fork Detection:** Clients detect server equivocation.
- **Fork Detection:** Clients detect server equivocation.

$$\text{Commit Hash} = \text{SHA256}(\text{Data} + \text{Prev_Hash})$$



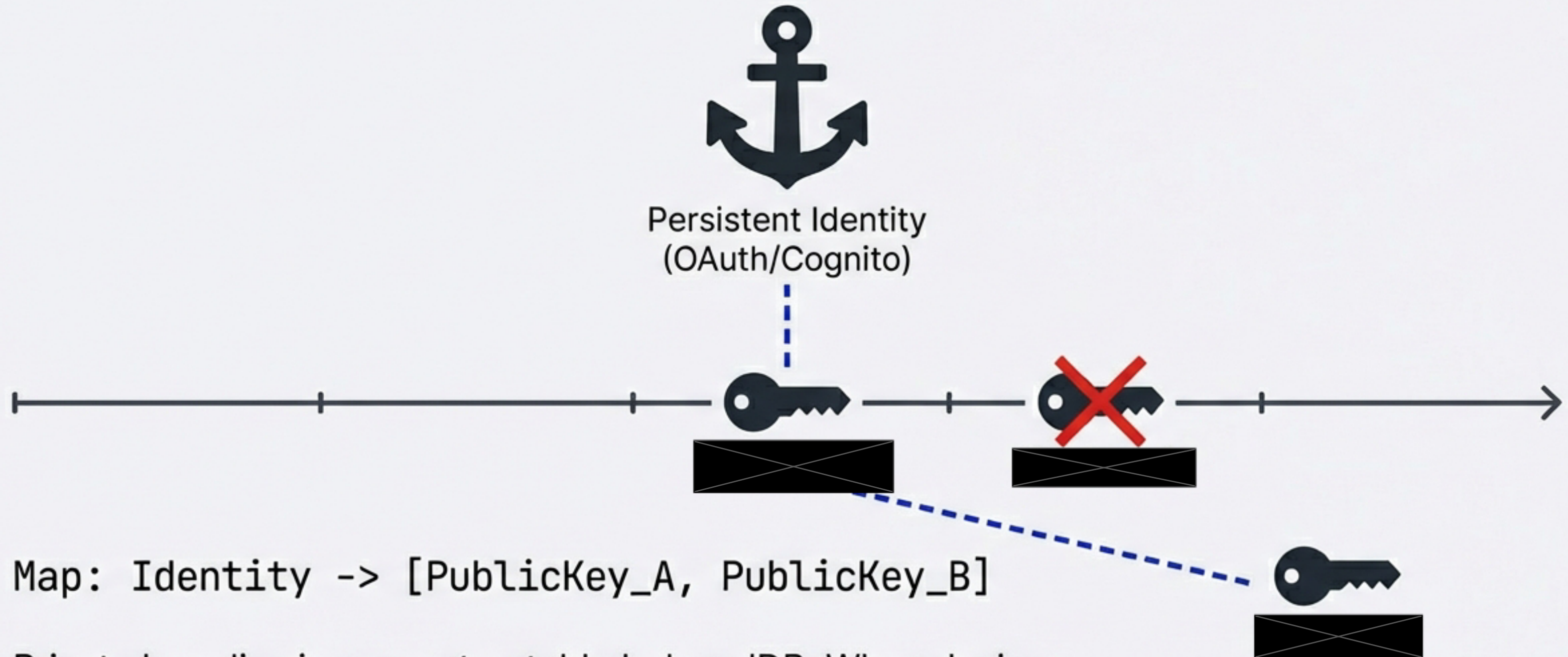
Every Change is Signed and Attributable



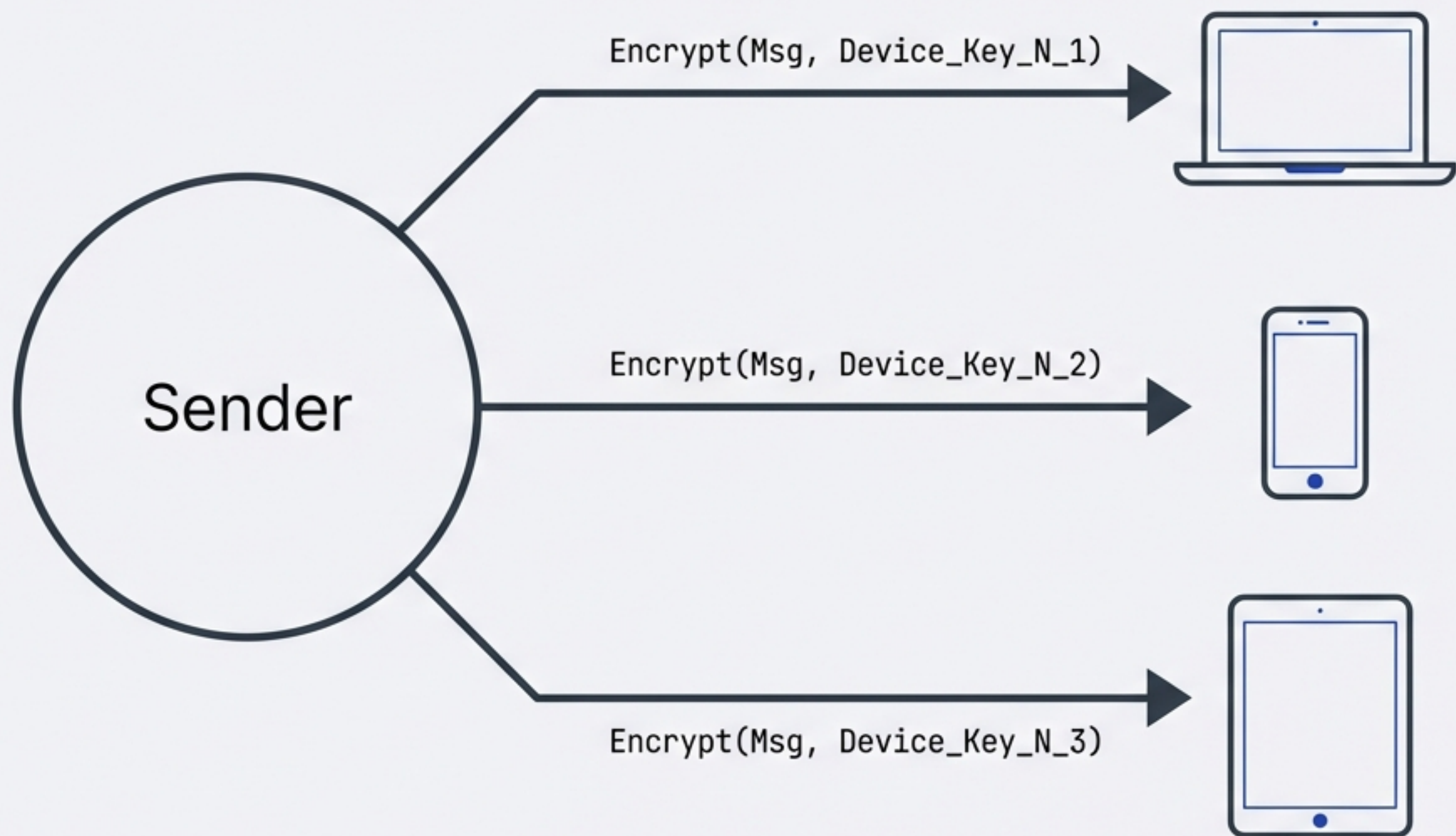
Scenarios

- Admin adds member → Signed by Admin Key.
- Server rotates key → Signed by Old Key.
- User rotates key → Signed by Identity Provider.

Identity Anchors the Ephemeral Key



Multi-Device State & Rotation



- **Model:** Follows Signal/PGP subkey architecture.
- **Rotation:** Not a replacement. A transactional "ADD_KEY" + "REVOKE_OLD_KEY" commit.
- **Result:** One identity, multiple valid decryption endpoints.

Revocation is Graph Maintenance

Before



Status: Trusted

After



Status: Untrusted

Traditional PKI

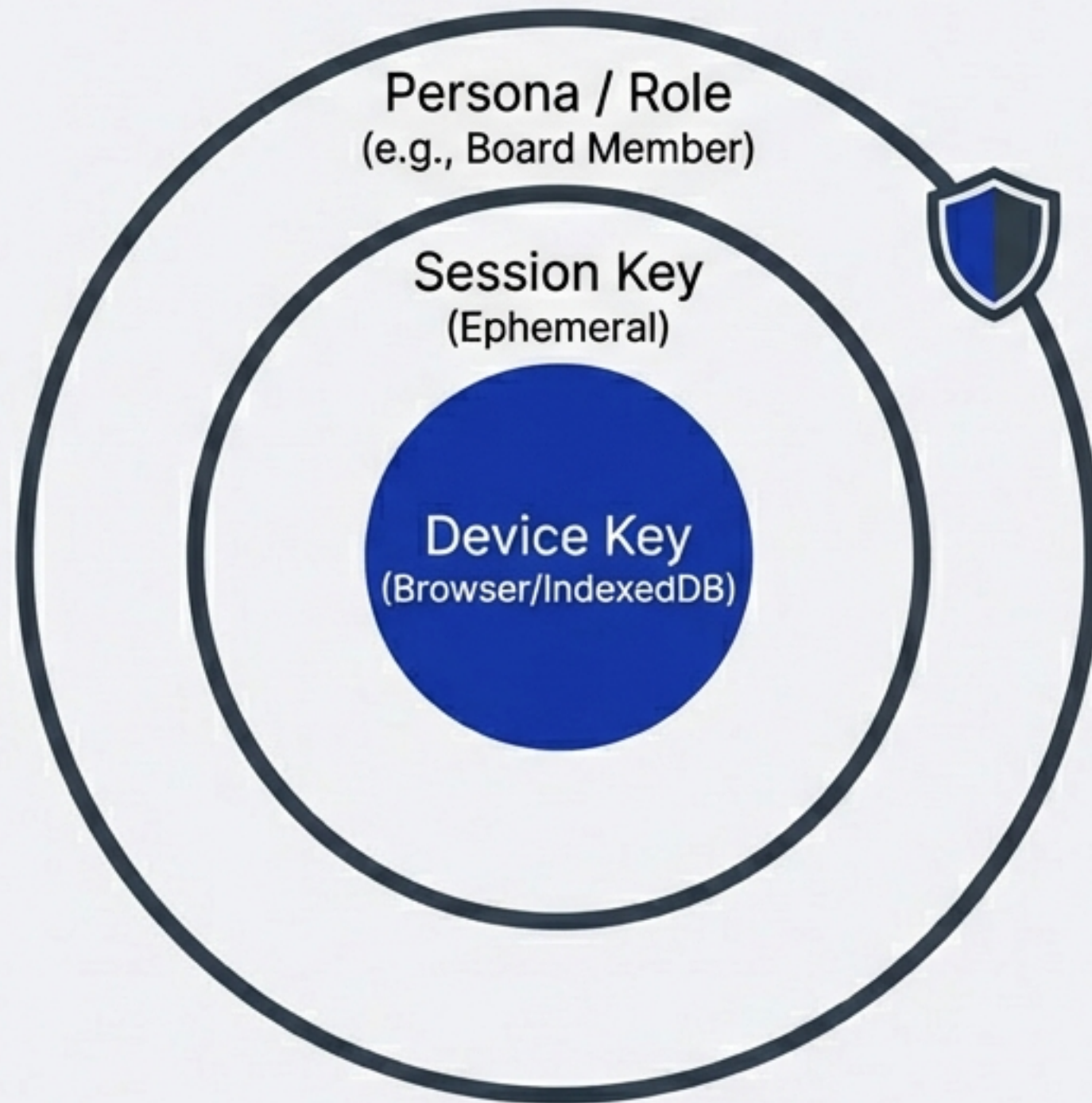
- Uses CRL/OCSP lists.
- High latency.
- Vulnerable to fail-open.

Graph Model

- Revocation is the absence of an edge.
- Instant propagation.
- Fails closed.

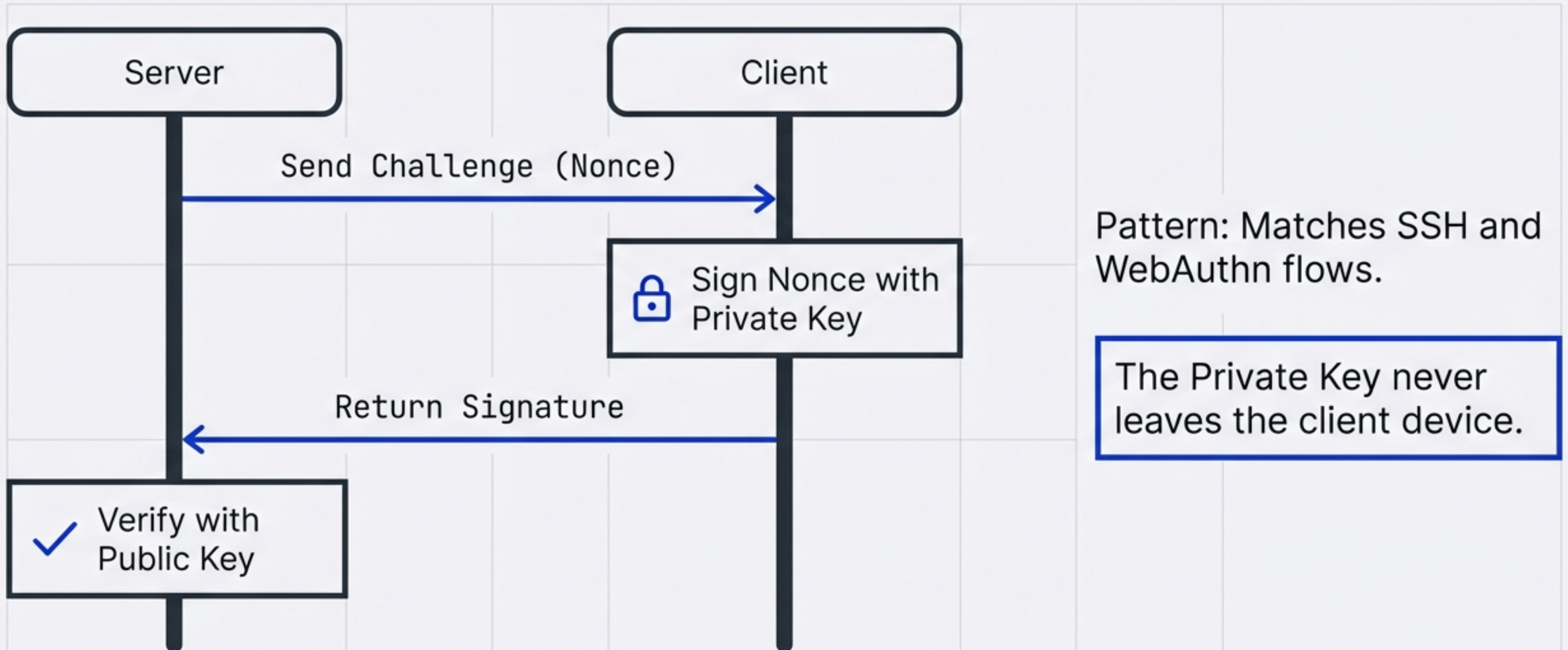
“If the graph doesn’t connect, trust does not exist.”

The Layered Identity Hierarchy

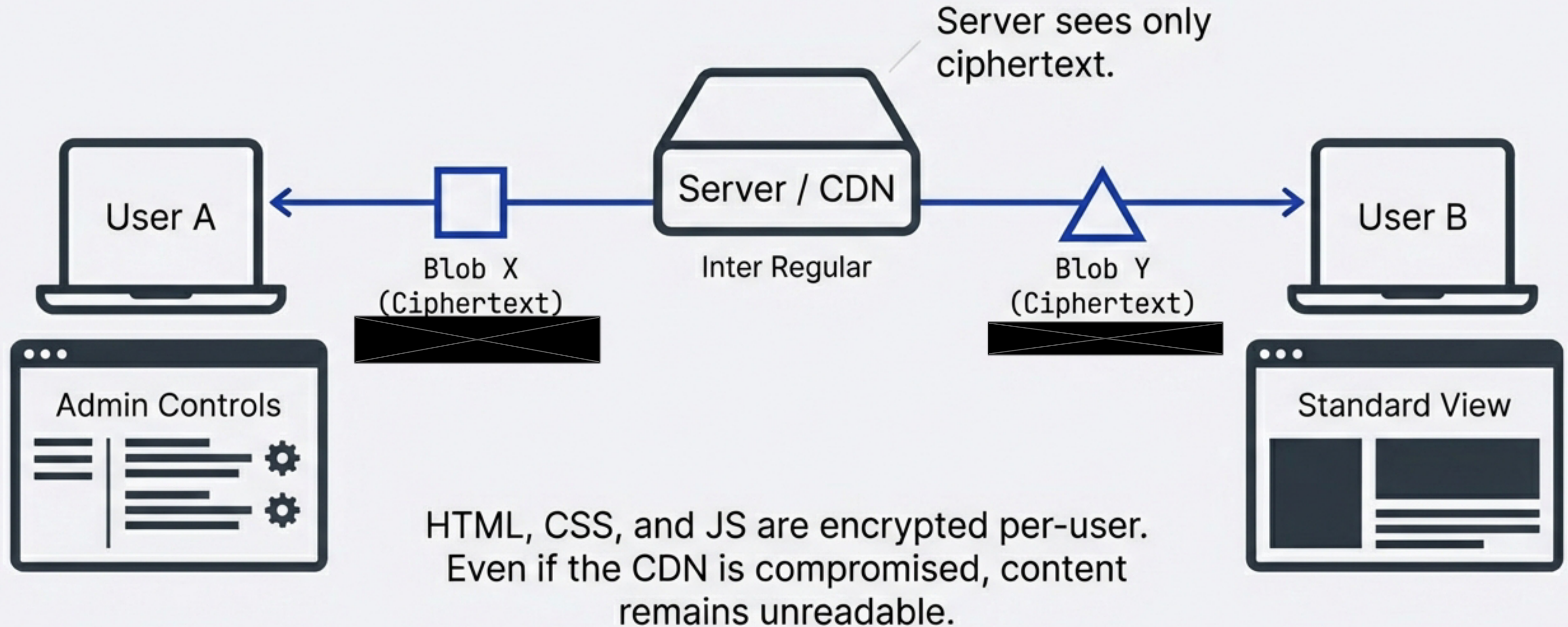


- Security is not binary; it is layered.
- Compromising a Session does not compromise the Admin Persona.
- Personas are cryptographically isolated.

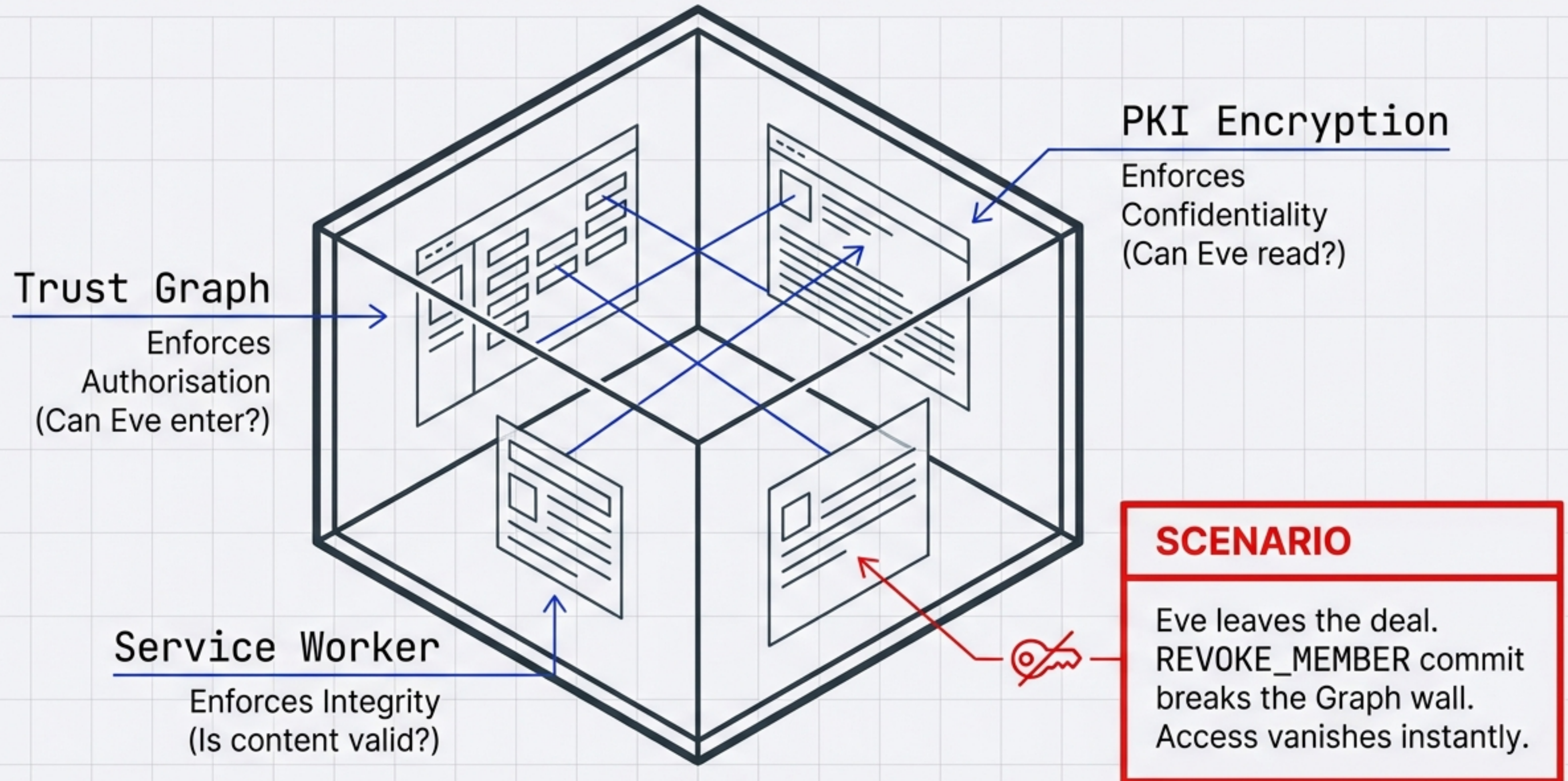
Challenge-Response Authentication



Personalised, Encrypted Web Delivery



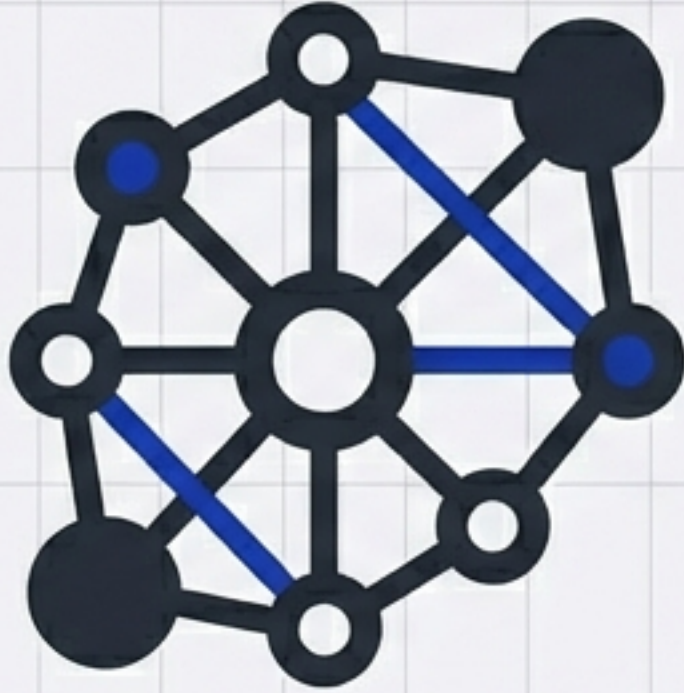
Architecture in Action: The Data Room



Engineering Priorities & Roadmap

Phase 1: Foundation (Now)	Phase 2: Integrity (Next)	Phase 3: Scale (Later)
Design Trust Graph Schema (Issues-FS)	Git-Style Hash Chaining	Key Rotation Flows
Key Discovery UI	Challenge-Response Protocol	Encrypted Page Prototypes
	Equivocation Attack Review	Research: Multi-user Secure Pods

The Three Pillars of Trust



1. Graph-Based Discovery



2. Hash-Chained Integrity



3. Layered Identity

A trust model where the server knows nothing, and the client verifies everything.